



Progress Update 2: Segfault Slayer

Schmölz Lukas, Schöpf Emanuel, **Schweigl Dominik**

Game Overview

Game Idea:

- Metroidvania-style 2D side-scroller
- Player fights programming concepts and bugs (mainly C++ related)
- Playful, Pixel-theme (e.g. ASCII-Art for textures)

World Design:

- First Area: Hardware with Logic Operators
- Second Area: Programming with loops and other concepts

Previous Goals

- Proper rendering
- Proper physics
- Additional content (player, enemies, health, rooms, ...)

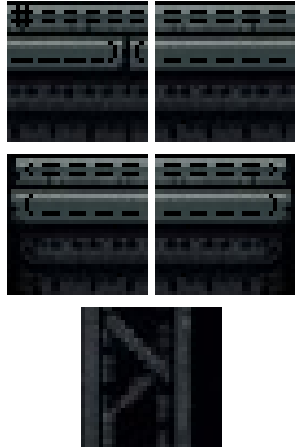
Textures & Rooms

Improved Tile Textures

- Pixel art tiles — ongoing, more tiles being added
- Pixellab explored for tile creation

Multiple Rooms

- World split into separate room files
- Room switching via coordinate check
- To be replaced with teleport tiles



Asset Manager

Centralised Texture Loading

- Singleton `AssetManager` owns all textures
- `TextureAsset` enum as type-safe asset keys
- No `loadFromFile` calls in entity code
- Textures loaded once, shared by `const` reference

```
AssetManager::getInstance()  
    .getTexture(PLAYER_HEAD);
```

Player — Sprite System

Three-Layer Sprite Architecture

- **Lower body** — motion animations
- **Upper body** — motion animations / attack animations
- **Head** — hat / no-hat (managed by Player)

Per-Frame Offset System

- Each state exposes `getHeadOffset()` and `getUpperBodyOffset()` per frame
- Enables composition across all states



Player — Hat Throw Ability

New Player Ability

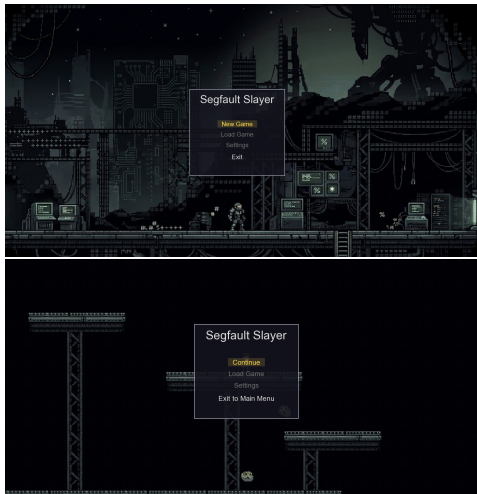
- Player throws hat as a projectile
- Head switches to hatless sprite once hat is taken off, checked via `isHatOnHead()`
- Separate `HatProjectile` entity
- Throw speed and distance adjusted based on player speed



Menu System

Stack-Based Scene Architecture

- Scene interface pushed/popped on a stack
- Game logic moved to GameScene
- MenuScene base class for menus
- **Main Menu** with background image
- **Pause Menu** pushed on top of GameScene



Demo & Code Review

Next Steps

- Player/Enemy Health System
- More Enemies / Boss
- Keyboard Binding Management
- Entities spawns in the Map
- Game Saves